

DFA REAL

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#include <stdio.h>
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#include <string.h>
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#include <ctype.h>
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```
char stat_table[6][4][10] = {  
    {"stat", "letter", "digit", "."},  
    {"start", "id", "int", "error"},  
    {"id", "id", "id", "error"},  
    {"int", "error", "int", "s"},  
    {"s", "error", "real", "error"},  
    {"real", "error", "real", "error"},  
};
```

```
int main() {  
    char input[20], column_stat[10], current_stat[10], next_stat[10];  
    char ch, choice;  
    int error, i, c, r, len;  
  
    do {  
        printf("Enter identifier: ");  
        scanf("%s", input);  
        len = strlen(input);  
        strcpy(current_stat, "start");  
        error = 0; // Reset the error before each input  
  
        for (i = 0; i < len && !error; i++) {
```

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ch = input[i];

if (isalpha(ch)) {
    strcpy(column_stat, "letter");
} else if (isdigit(ch)) {
    strcpy(column_stat, "digit");
} else if (ch == '.') {
    strcpy(column_stat, ".");
} else {
    strcpy(next_stat, "error");
    error = 1;
    break; // Exit the loop immediately when encountering an error
}

for (r = 1; r < 6; r++) {
    if (strcmp(stat_table[r][0], current_stat) == 0) {
        for (c = 1; c < 4; c++) {
            if (strcmp(stat_table[0][c], column_stat) == 0) {
                strcpy(next_stat, stat_table[r][c]);
                break;
            }
        }
    }

    if (strcmp(next_stat, "error") == 0) {
        error = 1; // Set error to 1 to break out of the loop
        break;
    }
}

```

```

        printf("%s %c %s\n", current_stat, ch, next_stat);
        strcpy(current_stat, next_stat);
        break;
    }
}
}

```

```

if (error) {
    printf("\nInvalid Token");
} else {
    printf("\nValid");
}

```

```

if (strcmp(current_stat, "id") == 0) {
    printf("\nIt is an identifier");
} else if (strcmp(current_stat, "int") == 0) {
    printf("\nIt is an integer");
} else if (strcmp(current_stat, "real") == 0) {
    printf("\nIt is a real");
}
}

```

```

printf("\n\nDo you want to continue? (enter 'y' for yes and 'n' for no): ");
scanf(" %c", &choice);
} while (choice != 'n');

return 0;

```

